Geo-ViT: A Geometry-Aware Vision Transformer Framework for Underwater Image Enhancement

Thallapally Harika[1] **and Kankanala Srinivas**[1,*]

[1]School of Electronics Engineering, VIT-AP University, Inavolu, Beside AP Secretariat, Amaravati, AP 522237, India
[*]Corresponding author: Kankanala Srinivas (srinivas.kankanala@vitap.ac.in)

ABSTRACT

The underwater image enhancement (UIE) process is heavily restricted by depth-dependent color casting and light scattering issues. Although the performance of deep learning-based UIE has been remarkably improved recently, many previous solutions have failed to successfully handle both the problems of preserving local texture information and global color consistency under various degraded underwater environments. To overcome the challenges mentioned above, Geo-ViT is introduced as a novel geometry-aware UIE framework. The network architecture adopts a symmetric U-Net structure along with a novel Geometry Encoder which employs the learnable Sobel filter to capture geometric prior knowledge. By combining the geometric feature map and the first stage of encoders via an additive feature fusion approach, the geometric information can be preserved during the downsampling process. To balance local texture recovery with global color consistency, the architecture features a Hybrid Bottleneck comprising interleaved Gated Local Context Block (GLCB) and Vision Transformer (ViT) blocks. While ViT parts encode global dependencies for the same color correction task, the GLCB modules propagate important spatial information while filtering out underwater noises. Further stabilization of the restoration model is done by a global residual connection to learn enhancement residuals rather than performing direct mapping tasks on the image. The effectiveness of the suggested method is measured against several benchmarks such as UIEB, EUVP (UWSCENES and UWIMAGENET), LSUI, and U45. Experimental analysis shows that the combination of geometric priors with hierarchical encoding and Charbonnier Perceptual loss function contributes to the best-in-class reconstruction ability and perceptual quality provided by the suggested model for underwater images.

Keywords: Underwater Image Enhancement (UIE), Computer Vision and Graphics, Vision Transformer (ViT), Geometry-Aware Learning, Gated Local Context Block (GLCB), Hybrid Bottleneck.

1 Introduction

Underwater image processing plays a crucial role in today's advanced marine technology for accomplishing tasks such as AUV navigation [1], monitoring of coral reefs, and inspection of subsea structures. The absorption of light and scattering caused due to the presence of water particles result in degraded underwater images [2]. This gives rise to inconsistent color cast and poor visibility, adversely affecting the performance of high-level vision tasks. The generation of high-quality underwater images is extremely important in understanding the marine ecosystem and performing various activities related to the underwater world, including underwater 3-D shape reconstruction, underwater object detection and recognition, underwater image semantic segmentation, underwater observations, and detection of submarine buried pipelines [3].

However, due to wavelength and distance-dependent light absorption and scattering, captured underwater images usually suffer from severe quality degradation problems, such as low contrast, color casts, and blurred details, these issues impose significant limitations on subsequent visual perception analysis and underwater world understanding [4]. Therefore, an effective underwater image enhancement (UIE) solution is expected and essential.

In recent years, the rapid growth of deep learning (DL) [5] has significantly impacted the field of UIE [6]. Learning based techniques are gaining much popularity due to their amazing performance and driving the evolution of UIE [7]. Some of the key difficulties remain unsolved although great efforts have been made towards building complex datasets and high performance learning frameworks. One of them is the use of supervised learning that requires large amounts of annotated data for training. Yet it is not an easy task to collect a lot of data from different underwater scenes by means of classical optical or polarized imaging equipment. The absence of water-free reference images further complicates the process of model training, which impacts the final performance of the model itself [8]. All the same, the existing approaches of UIE currently struggle with a fundamental problem of ensuring the simultaneous preservation of structural details and global color consistency. While some UIE algorithms emphasize convolutional feature extraction and others utilize transformers to capture global dependencies, none of them manages to combine both. Furthermore, the absence of explicit geometric priors limits the ability of these models to retain fine structural information in severely degraded regions.

High-quality images acquired via traditional UIE methods can serve as valuable references to enhance the performance of neural networks in high-level vision tasks [9]. We contend that learning-based methods will play a pivotal role in the future of UIE. Currently, it appears that entirely replacing traditional UIE with a purely learning-based approach is impractical, as many learning-based frameworks are still constructed by integrating features extracted using measures from traditional UIE methods [10]. Therefore, advancements in traditional UIE methods may offer novel insights and benchmarks for enhancing the metrics and performance of learning-based frameworks, especially in enhancing the quality of reference annotations.

Additionally, we observe that most existing deep UIE models [11], [12] heavily depend on the engineering knowledge (such as multi-scale fusion, degradation separation, joint iterations, Retinex decomposition) in their architectures, making the performance of such models significantly dependent on the size and quality of the training datasets and not very robust to real-life underwater environment. On the contrary, many domain specific knowledge about the underwater images is actually hidden within the degradation effects, and using them can help the model to learn more discriminative features and reconstruct better images, as well as increase the generalization capabilities of the model. There are two examples of such degradation effects: raw images with the degradation and corresponding three priors (background light, water type, and scene depth). Background light of the underwater image contains some distortion information regarding color casts, which can be used for correcting colors. With the use of the water type prior, which contains the information about water bodies, the trained model will become more robust to different water bodies. The scene depth reflects the degradation level of underwater images.

This paper introduces a novel hybrid framework, termed Geo-ViT (Geometry-Aware Vision Transformer), for UIE. The major contributions of this paper can be summarized as follows:

- Geometry-aware feature extraction using differentiable Sobel operators for preserving structural details and edge information.
- A hybrid bottleneck combining GLCB and Vision Transformer blocks to jointly learn local textures and global contextual representations.
- A gated feature extraction mechanism for suppressing underwater noise while preserving spatial information.
- Global residual learning and a hybrid loss function for improved reconstruction quality.
- Superior performance compared with state-of-the-art methods on UIEB, EUVP, LSUI, and U45 benchmark datasets.

The remaining sections of this paper are as follows. Section II gives an overview of the related works in UIE, including traditional CNN-based methods and transformer-based methods. Section III provides the architecture of the Geo-ViT method. Section IV gives details of experiments conducted in terms of data preparation, evaluation, and results. Section V provides conclusion and future work in this domain.

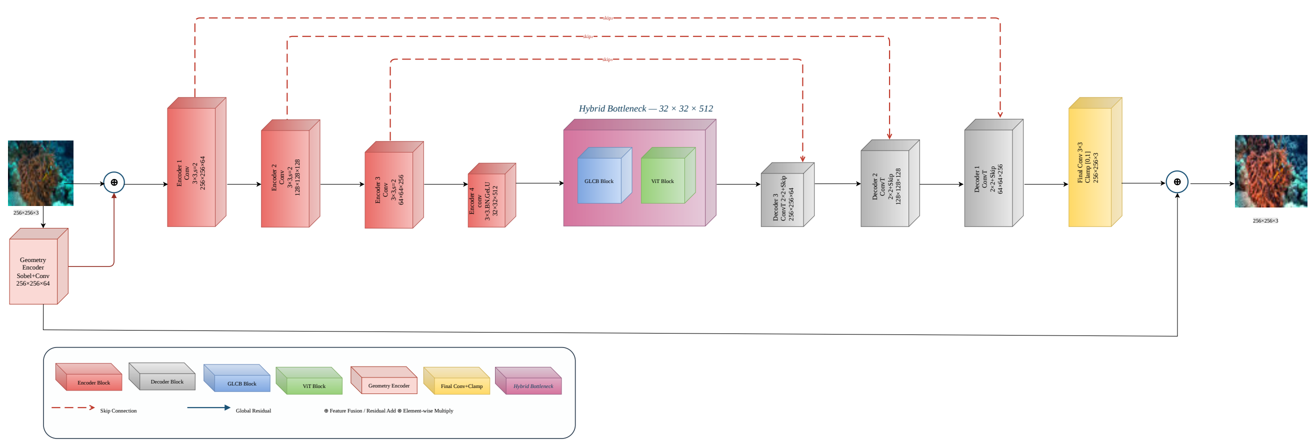

Figure 1. Detailed architecture of the proposed Geo-ViT network. The model utilizes a symmetric U-Net structure with a Geometry Encoder to inject structural priors, and a Hybrid Bottleneck (GLCB & ViT) to balance local context with long-range dependencies.

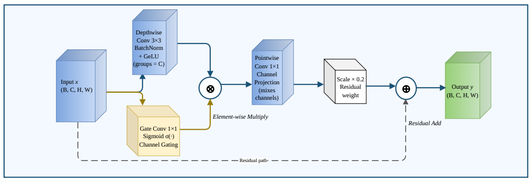

Figure 2. The Gated Local-Context Block (GLCB) extracts local spatial features while dynamically suppressing noise.

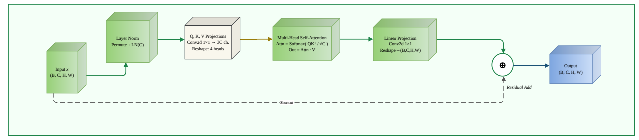

Figure 3. The standard ViT Block captures long-range global dependencies.

2 Related Work

The architecture landscape for UIE has quickly transitioned from physics-based equations to accurate deep-learning methods. This section provides a critical analysis of the evolution from initial CNNs to vision transformers, including a discussion on the development of hybrid models with improved geometrical and gating priors.

2.1 Traditional Frameworks and Convolutional Neural Networks

The early efforts for in UIE were very much focused on the concept of inverting the physical Underwater Image Formation Model (UIFM). The algorithms typically depended on making certain statistical assumptions about the scene, most often by employing the Dark Channel Prior (DCP) technique [13] to model transmission medium and address the issue of turbid water. This does not work well for scenes having non-uniform lighting conditions.

In order to address these limitations posed by determinism, the supervised learning based CNN model was the dominant paradigm adopted for solving the problem. Certain architectural paradigms such as UWCNN [14] and Shallow-UWNet [15] have proposed certain custom multi scale filter banks specifically designed to cater to various types of water masses in the ocean. Following this, WWUIE [16] algorithm was introduced, which used wavelets for feature decomposition along with white balance correction algorithms.

However, one of the main drawbacks of the classical convolutional structures lies in their limited receptive field. Due to spatial limitations, it becomes impossible for pure CNN algorithms to build long range dependencies, thus making it impossible to correct long range color distortions and maintaining global contrast in extremely degraded benthic environment.

2.2 Vision Transformers and Contemporary Attentive Paradigms

Vision Transformers (ViT) revolutionized low-level image processing as they substituted spatially localized convolution operators with Multi-Head Self-Attention (MHSA) approaches. As regards underwater environments, architectures such as UWFormer [17] efficiently implemented window-tokenized MHSA in order to construct dependency graphs at the global level, which was crucial for addressing issues associated with non-uniform attenuation and global color casts. Along these lines, the approach of DDFormer [18] introduced the use of a dual-domain matrix to perform concurrent operations on latent tokens within both spatial and coordinate domains.

However, recent developments have focused on tuning transformer weights more efficiently in order to understand the complex optical phenomena in underwater environments. For instance, the Waterformer model was developed using light attenuation vectors incorporated into its self-attention mechanisms in order to address backscattering issues. However, modern research has progressed towards decoupling transformers in the context of a certain domain. One prominent example is the switchable attention and transformer based multiscale disentanglement method introduced by Wang [19].

However, despite their good ability in global color balance, there are still some architectural weaknesses present in purely transformer models. First, the quadratic computational complexity regarding token length renders their use ineffective on high-resolution input on limited resources. More importantly, they do not have any inductive biases for localized pixel neighborhoods and thus may end up creating blur and losing edges in highly textured regions.

2.3 Geometric Priors and Edge-Aware Networks

The latest research work on UIE has focused on the significance of including edge information into the process of feature extraction to maintain structure information which is lost due to the processes of scattering and downsampling. Wavelet based methods for edge enhancement, for instance, show how the use of boundary information in higher frequencies can lead to better restoration of textures and structures in the resultant images obtained through UIE techniques [20].

Moreover, the preservation of directional gradient helps the latest restoration techniques preserve excellent spatial consistency and better object boundaries during the restoration process. It mitigates the tendency for image oversmoothness that is typically associated with CNN-based techniques and enhances the restoration of crucial visual elements. Based on the above discussion, a recent technique has considered two different branches that complement each other in incorporating both physical properties and edge features [21].

2.4 Gated Context Mechanisms and Advanced Hybrid Systems

In order to have a balanced performance in terms of local frequency sharpness and global chromatic accuracy, current work on UIE focuses greatly on hybrid models that combine convolutional features with transformers for global feature processing. Prior hybrid models, like P2CNet [22] that guided cluster by clusters using convolution and encoder of transformer, inevitably introduced a large feature alignment challenge.

The state-of-the-art method currently eliminates the limitation by incorporating the use of Gated Local Context Block (GLCB) and dynamic channel gating. Having been introduced in general vision models and later applied to image restoration challenges, gated convolutions apply element-wise activations in the form of sigmoid function that determines the reliability of the features. In an environment where particles are numerous causing backscatter noise, gated layers computationally eliminate regions characterized by scattered noise and pass on intact regions of local spatial context.

Recently, however, the incorporation of such dynamic channels within multi-scale adaptive fusion networks has been achieved in modern frameworks [23]. While these highly commendable innovations have been made, there is yet much to be done since existing hybrid frameworks still face difficulties in establishing connections at both the geometry domain and the global attention domain simultaneously.

While most recent architectures emphasize the importance of gated noise filters or make use of abstract global transformers, no attention is paid to the equally important aspect of linking geometry edge mapping with spatially gated streams. This research gap is directly addressed by our suggested Geo-ViT model which combines the benefits of geometry encoder along with the hybrid bottleneck consisting of GLCBs and ViT blocks.

3 Proposed Methodology

3.1 Geo-ViT Model Structure

Architecture of the proposed method is defined within the framework of symmetric U-Net, which uses geometric information-based priors through transformer-convolutional bottleneck structure. It can be seen from Fig. 1 that degradation effects are successfully restored by applying texture-color corrections using the proposed model.

3.2 Geometry-Aware Feature Extraction

The recovery algorithm starts with an encoder architecture known as Geometry Encoder branch. While common encoders learn the features by themselves, this particular module uses learnable Sobel filters to calculate ∇_x and ∇_y of the original image I. The structural edge map G is derived as:

$$G = \sqrt{(\nabla_x I)^2 + (\nabla_y I)^2 + \varepsilon} \tag{1}$$

As defined in Eq. (1), where $\varepsilon = 10^{-6}$ for numerical stability. This geometric feature map is integrated into the primary latent space via additive feature fusion at the output of the first encoder stage

$$F_{geo} = F_{enc}^{(1)} + G \tag{2}$$

In Eq. (2), $F_{enc}^{(1)}$ refers to the feature map produced by the first stage of encoding, while G refers to the geometry-aware edge feature map calculated from Eq. (1). Geometry-aware features are combined with the encoder features using element-wise addition in order to enhance structural information.

The projected geometry features are transformed into higher-dimensional latent representations through convolutional projection, which is formulated as

$$F_G = W_G * G \tag{3}$$

As defined in Eq. (3), where W_G denotes the learnable convolution kernel and G represents the geometry-aware edge feature map.

3.3 Hierarchical Encoder-Decoder Structure

The network architecture is based on an encoder-decoder architecture, which comprises of a hierarchical encoder consisting of four layers and a corresponding decoder comprising of three layers. During the process of downsampling, each layer E_n makes use of the 3×3 convolutions with GeLU activations to gradually reduce spatial resolution from 256×256 to 32×32.

$$D_n = \text{Dec}_n(x) + E_m \tag{4}$$

As defined in (4), where D_n represents the decoder feature at stage n and E_m is the corresponding encoder feature.

The hierarchical encoder extracts feature representations at different scales using convolutional operations followed by GeLU activation, which can be expressed as

$$E_n = \phi(W_n * E_{n-1}) \tag{5}$$

As defined in Eq. (5), where W_n represents the convolution kernel and $\phi(\cdot)$ denotes the GeLU activation function.

3.4 Hybrid Bottleneck (GLCB & ViT)

As shown in Fig. 2 and Fig. 3, the hybrid bottleneck integrates both local and global feature extraction. At the lowest resolution (32×32), the model employs a Hybrid Bottleneck consisting of interleaved blocks.

3.4.1 Gated Local Context Block (GLCB)

The GLCB utilizes a gating mechanism to filter underwater artifacts. Given an input x, as defined in (6),

$$g = \sigma(W_g * x) \tag{6}$$

The final output y_{GLCB} is computed using a residual gating operation:

$$y_{GLCB} = x + \alpha \cdot (W_{pw} * (\text{GELU}(\text{Norm}(\text{DWConv}(x))) \odot g)) \tag{7}$$

As defined in (7), where $\odot$ denotes element-wise multiplication and σ is the sigmoid function.

3.4.2 Vision Transformer (ViT)

As defined in (8), the ViT blocks employ multi-head self-attention (MHSA) to capture long-range dependencies for uniform color correction.

$$\text{Attention}(Q,K,V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{8}$$

The local features extracted by the GLCB are subsequently combined with the global contextual representations generated by the ViT block.

$$F_{HB} = F_{GLCB} + F_{ViT} \tag{9}$$

As defined in Eq. (9), F_{GLCB} and F_{ViT} denote the feature representations generated by the Gated Local Context Block and Vision Transformer blocks, respectively. The hybrid bottleneck combines local structural information with global contextual dependencies, enabling effective restoration of underwater degradations while preserving fine-grained details.

3.5 Global Residual Learning

As defined in (10), the network adopts a Global Residual strategy. The final enhanced image I_{enh} is formulated as:

$$I_{enh} = \text{Clamp}(I_{in} + \text{Net}(I_{in})) \tag{10}$$

3.6 Loss Functions and Optimization

To achieve high PSNR and perceptual quality, we optimize the network using a hybrid loss function $\mathscr{L}_{total}$:

$$\mathscr{L}_{Char} = \frac{1}{N}\sum\sqrt{\|\hat{I} - I_{GT}\|^2 + \eta^2} \tag{11}$$

$$\mathscr{L}_{total} = \mathscr{L}_{Char} + \lambda \mathscr{L}_{VGG} \tag{12}$$

The Charbonnier loss in Equation (11) ensures that there is reliable reconstruction at the pixel level, whereas the total loss in Equation (12) incorporates perceptual consistency. In this context, $\mathscr{L}_{Char}$ refers to the Charbonnier loss, and $\mathscr{L}_{VGG}$ stands for the perceptual.

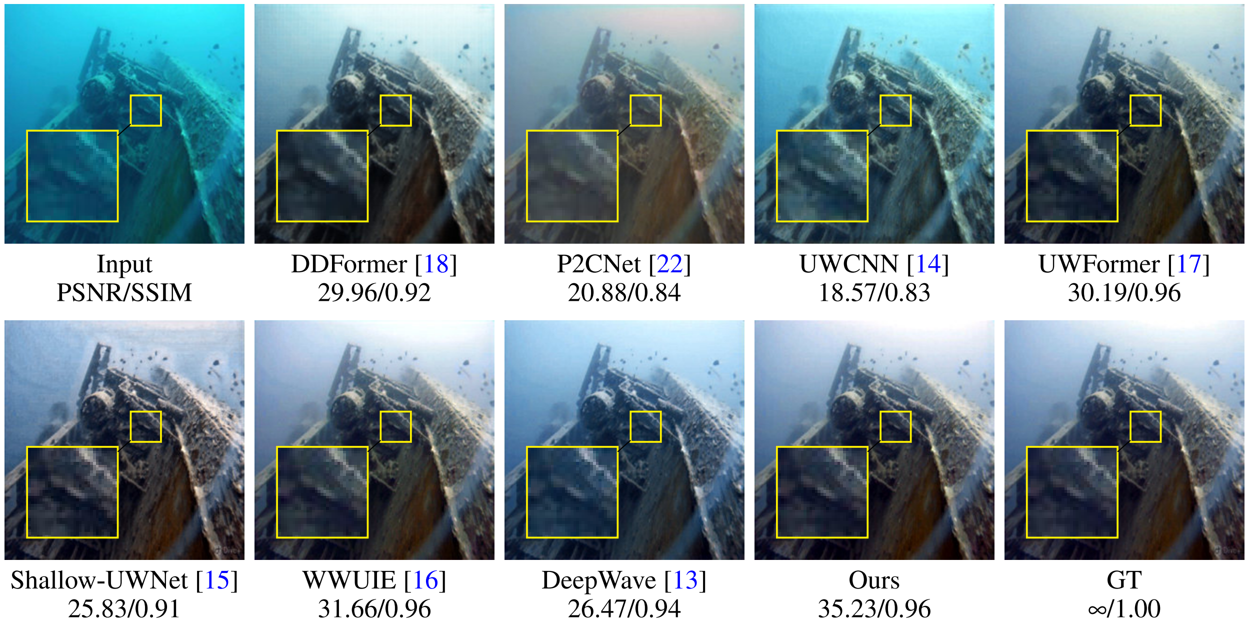

Figure 4. Visual comparison on Image 0643 from the UIEB dataset. Yellow boxes indicate enlarged local regions. The proposed Geo-ViT produces sharper edges and more faithful color restoration than the competing methods.

4 Experiments and Discussion

4.1 Experimental Setup

This section presents a comprehensive evaluation of the proposed Geo-ViT framework. All experiments are conducted to validate the effectiveness of the integrated geometry-aware architecture and the hybrid loss design in terms of both quantitative metrics and qualitative visual fidelity.

4.2 Datasets

Experiments were performed using four underwater image datasets such as UIEB [24], LSUI [26], and two sub-datasets of the EUVP [25] dataset, which are UWSCENES and UWIMAGENET, for testing the effectiveness and generalization capacity of our proposed enhancement framework. These datasets cover different types of underwater scenes with variable lighting conditions, distorted colors, and hazy effects. LSUI contains a total of 4,279 image pairs obtained from various underwater conditions with varying degrees of degradation. As per the traditional protocol, the dataset is split into 3,423 images for training and 856 images for testing purposes. It can be seen that with its diverse content and large size, the LSUI dataset can effectively test the robustness of any UIE technique. Additionally, the U45 dataset [27] was also utilized to perform qualitative and no-reference evaluations. The U45 dataset consists of 45 underwater images collected from different underwater environments and suffering from color cast, low contrast, scattering, and haze problems. No ground truth images are provided along with the U45 dataset. This is why, using all 45 images, only a no-reference evaluation has been carried out, and the corresponding quality metric scores like UIQM, UCIQE, and NIQE are reported.

The detailed dataset statistics and corresponding training and testing splits are summarized in Table 1.

Table 1. Summary of datasets and data splits used for training and evaluation.

Dataset	Subset	Total Images	Training	Testing
UIEB[24]	–	804	562	242
EUVP[25]	UWSCENES	2,185	1,530	655
EUVP[25]	UWIMAGENET	3,700	2,590	1,110
LSUI[26]	–	4279	3,423	856
U45[27]	–	45	–	45

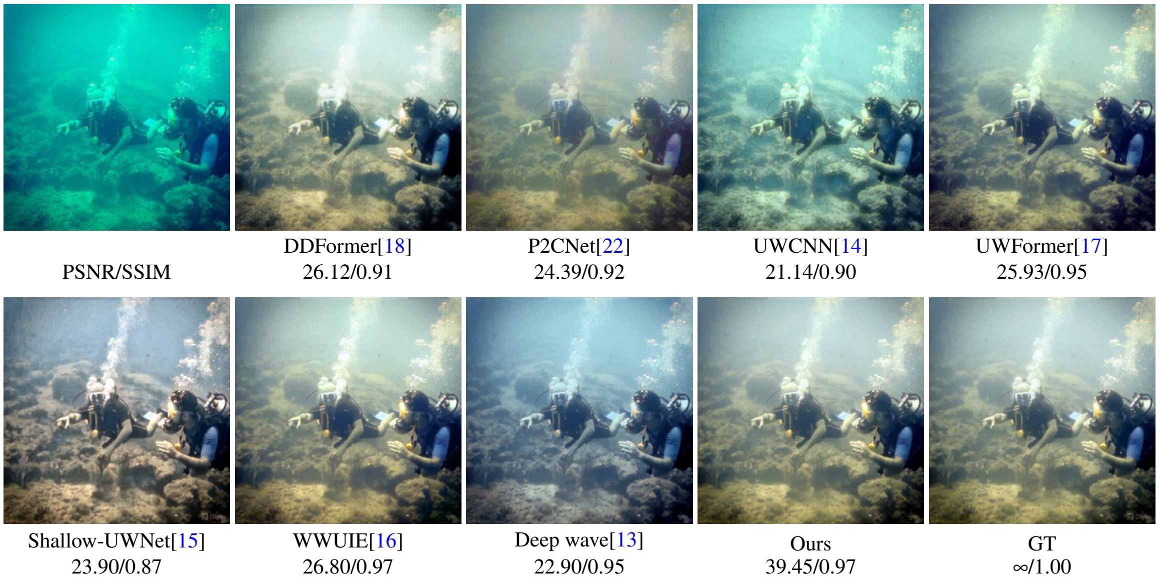

Figure 5. Visual comparison on Image 0743 from the UIEB dataset [24].

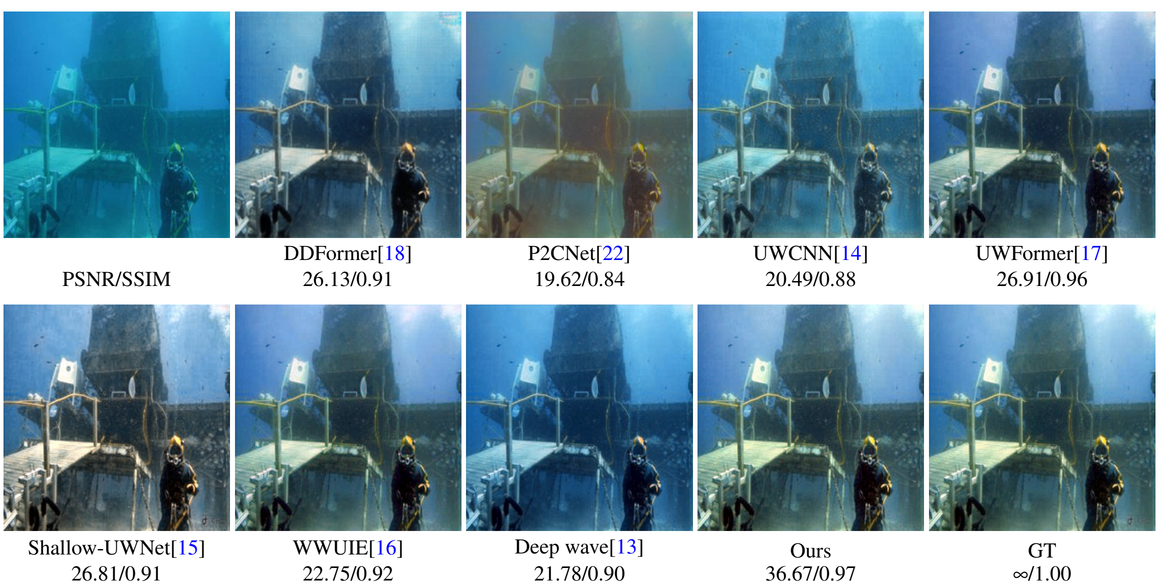

Figure 6. Visual comparison on Image 0711 from the UIEB dataset [24].

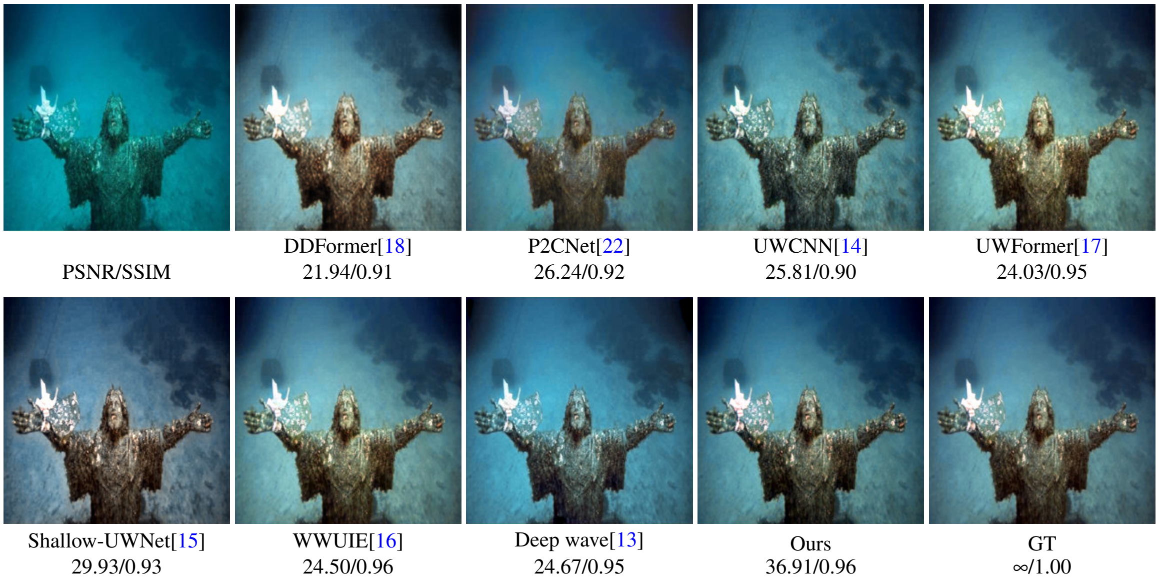

Figure 7. Visual comparison on Image 0686 from the UIEB dataset [24].

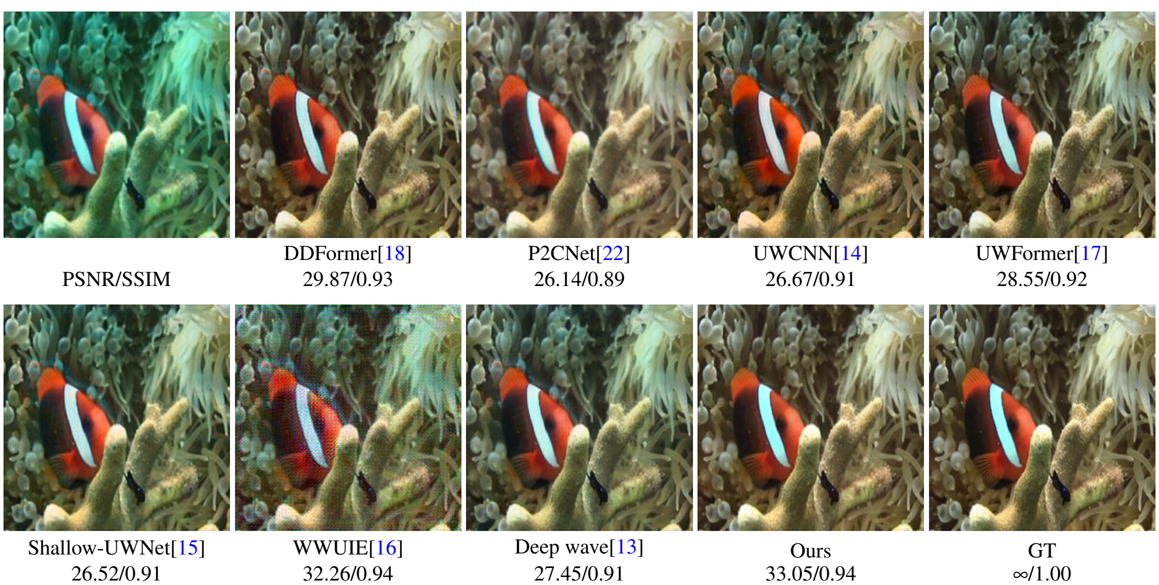

Figure 8. Visual comparison on Image 1645 from the EUVP dataset (UWSCENES subset) [25].

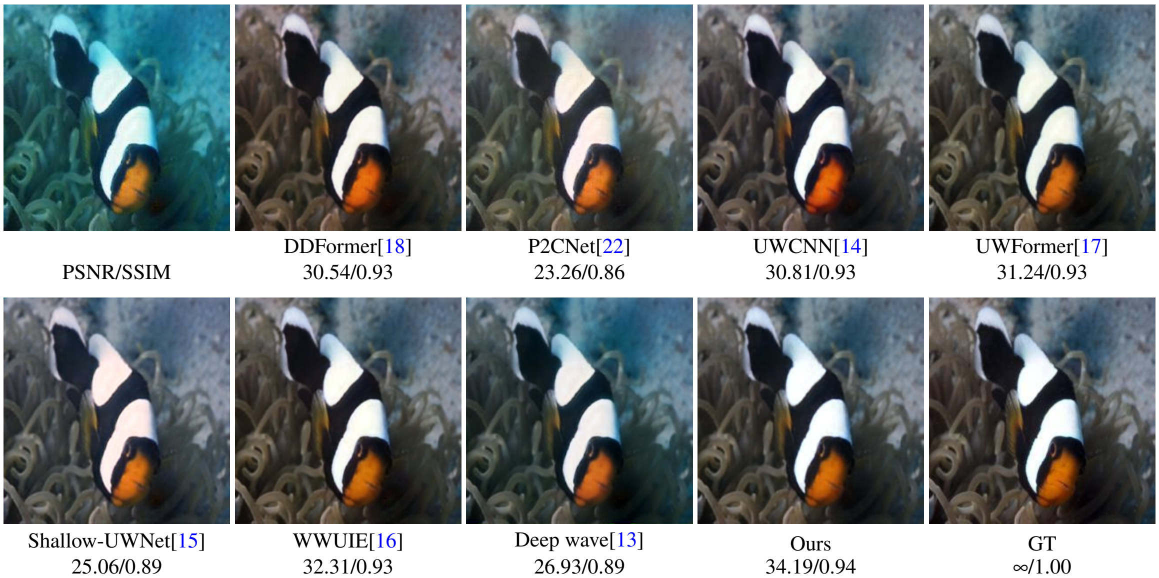

Figure 9. Visual comparison on Image 3000 from the EUVP dataset (UWIMAGENET subset) [25].

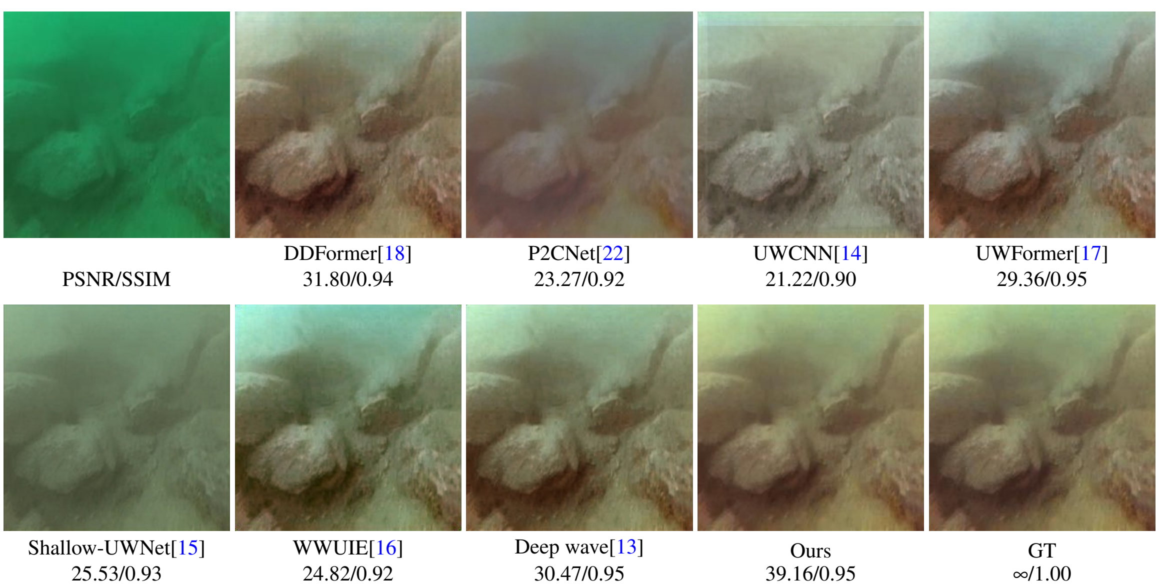

Figure 10. Visual comparison on Image 885 from the LSUI dataset [26].

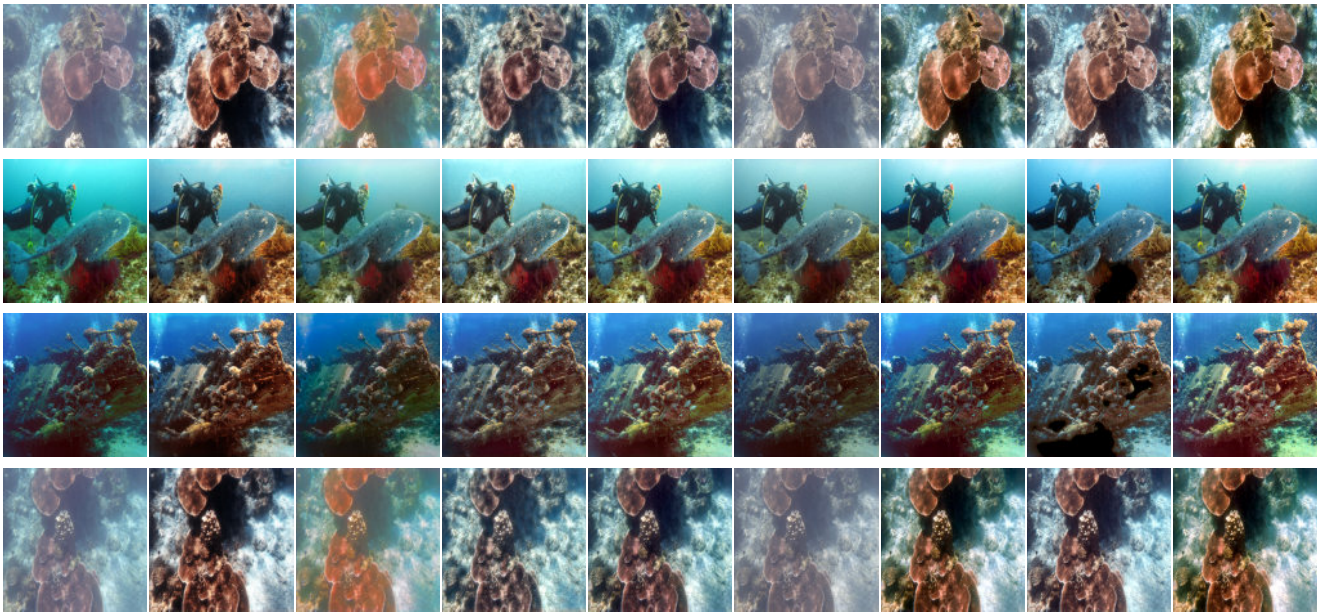

Figure 11. Visual comparison on representative images from the U45 dataset [27]. (a) Input, (b) DDFormer [18], (c) P2CNet [22], (d) UWCNN [14], (e) UWFormer [17], (f) Shallow-UWNet [15], (g) WWUIE [16], (h) DeepWave [13], and (i) Proposed Geo-ViT.

4.3 Implementation Details

The details of the proposed Geo-ViT method in terms of the implementation and training are provided in Table 2. Image input sizes are set at 256×256 pixels, while a batch size of 4 is considered in the training process. The AdamW algorithm with the momentum values of $\beta_1 = 0.9$ and $\beta_2 = 0.999$, weight decay rate 1×10^{-2}, and learning rate 2×10^{-4} is utilized. To guarantee convergence, a cosine annealing learning rate scheduler is implemented for 200 epochs. For training, the weighted sum of Charbonnier and perceptual loss functions is considered, with a weight value of 0.1. Experiments are carried out using an NVIDIA Tesla T4 GPU.

4.4 Evaluation Metrics

To comprehensively evaluate the performance of the UIE framework, both full-reference and no-reference image quality metrics are employed. These metrics jointly assess pixel-level fidelity, structural preservation, perceptual quality, color restoration, and naturalness of the enhanced images.

Peak Signal to Noise Ratio (PSNR)[28] is a full-reference performance measure that compares the pixel-to-pixel similarity of enhanced images and corresponding ground-truth images. High PSNR scores indicate accurate reconstruction and low pixel-level distortion. PSNR is an appropriate measure for paired databases since reference images can be easily obtained.

Structural Similarity Index Measure (SSIM)[29] is another full-reference measure that assesses structural similarity of enhanced images and their corresponding references with regards to luminance, contrast and structure. High SSIM scores indicate good retention of image structures and strong perceptual similarities, which is in agreement with human vision perception.

The Underwater Image Quality Measure (UIQM)[30] is a no-reference metric that specifically aims to measure underwater images. It measures the quality of the image based on three components of perception: colorfulness, sharpness, and contrast. Higher values of UIQM correspond to better visual quality and higher underwater visibility, without any reference needed.

The Natural Image Quality Evaluator (NIQE)[31] is a no-reference metric which determines the quality of the image based on its natural scene statistics. This metric is unique since contrary to the rest, the lower the NIQE score is, the higher perceptual quality is achieved.

In general, the high scores in PSNR, SSIM, UIQM, and Underwater Color Image Quality Assessment (UCIQE)[32] parameters correspond to good enhancement performance, while the low NIQE scores correspond to the natural appearance of images. It is obvious that their combined use ensures unbiased and consistent assessment of UIE quality from the objective and perceptual viewpoints.

Table 2. Training Configuration and Implementation Settings for the Geo-ViT.

Parameter	Value / Setting
Deep Learning Framework	PyTorch 2.x
Model Architecture	Geo-ViT
Training Strategy	Supervised (End-to-End)
Input Resolution	256×256 pixels
Batch Size	4
Optimizer	AdamW
Momentum Parameters	$\beta_1 = 0.9, \beta_2 = 0.999$
Weight Decay	1×10^{-2}
Initial Learning Rate	2×10^{-4}
Learning Rate Schedule	Cosine Annealing (200 epochs)
Loss Function	Charbonnier + 0.1 $\times$ Perceptual
Training Hardware	NVIDIA Tesla T4 GPU

4.5 Qualitative Comparison

The findings of the qualitative comparisons are presented in Figures 5, 4, 6, 7, 8, 9, 10, and 11. The underwater images under consideration have many degradation factors such as color distortions, insufficient contrast, blurry textures, and poor visibility. Even though the performance of existing methods is promising to some extent, there are also several drawbacks worth noting. In particular, the results obtained using the UWCNN and P2CNet methods may have residual color issues and contrast deficiencies, while those obtained using DDFormer and UWFormer tend to be overly smooth, which obscures fine structures.

Fig. 5, Fig. 4, Fig. 6 and Fig. 7 show that the proposed Geo-ViT consistently provides better-looking underwater images in terms of better color restoration, better contrast and better structural detail preservation. Specifically, Fig. 4 shows a visual comparison for Image 0643 of UIEB dataset, where the locally enlarged areas show that the proposed method successfully captures finer textures and reconstructs object contours along with better color restoration in comparison to other methods. Unlike DDFormer, P2CNet, UWCNN, UWFormer, Shallow-UWNet, WWUIE and DeepWave, the proposed framework not only provides better details but also remains visually consistent with the ground truth. Moreover, the proposed framework successfully eliminates the underwater color casts.

Qualitative evaluations conducted using the UWSCENES and UWIMAGENET benchmarks, illustrated in Figs. 8 and 9, respectively, further validate the effectiveness and reliability of our framework in different underwater settings. Restored images show a better level of visibility and more realistic color reconstruction, especially for areas suffering from strong light attenuation and distortion. In addition, the contours of objects, structures of scenes, and texture information have been reconstructed more accurately than those obtained using other CNN and transformer models. The comparison through visualization on the LSUI dataset shown in Fig. 10 provides another confirmation of the efficiency of the introduced framework for UIE at scale. Specifically, the reconstructed images obtained using Geo-ViT demonstrate higher quality of reconstruction in terms of clarity, faithful color representation, and detail preservation compared to that of competing techniques. Moreover, the artifacts are minimized, and the results appear more consistent with the references.

For the U45 benchmark dataset, which does not have available ground truth images, the subjective assessment from Fig. 11 shows that the proposed model performs well in terms of visibility improvement, contrast enhancement, and color adjustment of underwater scenes. The obtained results show improved object shape perception, less color distortion, and better appearance of the images than that of the existing state-of-the-art models. From the visual comparisons of all the datasets, we can say that the Geo-ViT framework is a good trade-off for UIE performance.

4.6 Quantitative Comparison

The results of the quantitative comparison analysis are presented in Tables 3 through 7, wherein both full-reference and no-reference measures have been included.

In the case of UIEB data set, the proposed algorithm provides the best PSNR and SSIM values of 25.56 dB and 0.92, respectively, compared to existing methods based on CNNs and transformers. Moreover, perceptual quality is also comparable, as evidenced by the high UCIQE value of 0.62 and low NIQE value of 4.37.

Table 3. Quantitative comparison on UIEB dataset[24].

Method	PSNR↑	SSIM↑	UIQM↑	UCIQE↑	NIQE↓
UWCNN[14]	19.84	0.84	4.32	0.55	4.31
ShallowNet[15]	24.99	0.90	**4.76**	0.62	**4.07**
WWUIE[16]	25.38	0.92	4.35	0.61	4.30
Deep wave[13]	22.60	0.87	4.55	0.60	4.43
P2CNet[22]	18.89	0.77	3.94	0.54	4.58
UWFormer[17]	23.62	0.90	4.63	0.59	4.53
DDFormer[18]	22.93	0.86	4.44	0.61	5.34
Geo-ViT	**25.56**	**0.92**	4.26	**0.62**	4.37

Table 4. Quantitative comparison on UWSCENES dataset[25].

Method	PSNR↑	SSIM↑	UIQM↑	UCIQE↑	NIQE↓
UWCNN[14]	26.00	0.87	**4.69**	0.58	5.73
ShallowNet[15]	26.21	0.87	4.28	0.57	5.58
WWUIE[16]	20.41	0.43	4.32	**0.60**	12.91
Deep wave[13]	26.54	0.86	4.22	0.57	7.49
P2CNet[22]	24.39	0.83	4.31	0.57	5.57
UWFormer[17]	27.79	0.89	4.42	0.57	5.32
DDFormer[18]	27.91	0.88	4.58	0.58	5.91
Geo-ViT	**29.79**	**0.91**	4.62	0.58	**5.15**

On the UWSCENES dataset, our approach achieves the best performance in terms of PSNR (29.79 dB) and SSIM (0.91). This is shown in high-quality results for UIQM (4.62), UCIQE (0.58), and NIQE (5.15). It indicates that we obtain an optimal enhancement result without generating any artifacts. Furthermore, the resulting enhanced images have improved structural quality in various underwater scenes with challenging lighting and visibility conditions. While other approaches have similar perceptual metric scores, our approach ensures optimal results for both structural and perceptual qualities.

Similarly, on the UWIMAGENET dataset, the proposed method records the highest PSNR and SSIM values of 26.29 dB and 0.85, respectively, while achieving competitive UIQM (5.43), UCIQE (0.60), and NIQE (6.14) scores. These results demonstrate the effectiveness of the proposed framework in restoring image quality while preserving important structural and perceptual information under diverse underwater conditions.

The quantitative analysis for the LSUI dataset is shown in Table 6. It can be seen that the proposed model Geo-ViT has achieved the best PSNR of 28.62 dB with a competitive SSIM of 0.88. Additionally, it has acquired the smallest NIQE score of 4.63 among all the competing models, which implies that our model has better perceptual quality and naturalness of images. Though UWFormer has obtained a higher SSIM score of 0.89 and DDFormer acquires the highest UCIQE of 0.61, the proposed model outperforms the competing methods on reconstruction accuracy and visual quality.

However, for the U45 dataset, which lacks ground truth references, only no-reference methods are considered, as presented in Table 7. In the case of evaluation based on U45 dataset, the trained model using UIEB dataset is considered without any further fine tuning in order to evaluate its ability to generalize on unseen underwater images. The proposed approach obtains the highest UCIQE value of 0.62 along with competitive values of UIQM and NIQE of 4.29 and 4.06, respectively. However, although UWFormer obtains slightly better UIQM value of 4.47 and WWUIE obtains minimum NIQE value of 3.95, the proposed approach provides a good trade-off between contrast enhancement, color correction, and perceptual image quality.

In summary, the results across all datasets reveal that the proposed method consistently improves reconstruction accuracy while maintaining stable perceptual quality, thereby demonstrating its robustness and generalization capability under a wide range of underwater imaging conditions.

4.7 Ablation Study and Discussion

To analyze the effectiveness of each part of the model separately, an ablation study was carried out on the UIEB dataset, and the obtained results are presented in Table 8. The replacement of the complete hybrid bottleneck, comprising the GLCB and ViT blocks, with the traditional bottleneck causes a reduction in PSNR from 25.56 dB to 21.42 dB and a reduction in SSIM from 0.92 to 0.90. These results show that local context modeling along with global feature modeling plays an essential role

Table 5. Quantitative comparison on UWIMAGENET dataset[25].

Method	PSNR↑	SSIM↑	UIQM↑	UCIQE↑	NIQE↓
UWCNN[14]	24.80	0.83	5.33	0.59	6.47
ShallowNet[15]	23.72	0.81	5.14	0.57	6.25
WWUIE[16]	25.97	0.85	5.31	0.60	6.29
Deep wave[13]	24.50	0.81	5.09	0.59	8.23
P2CNet[22]	22.38	0.77	**5.69**	0.57	**5.11**
UWFormer[17]	25.18	0.82	5.25	0.59	6.20
DDFormer[18]	25.23	0.83	5.37	0.60	6.69
Geo-ViT	**26.29**	**0.85**	5.43	**0.60**	6.14

Table 6. Quantitative comparison on LSUI dataset[26].

Method	PSNR↑	SSIM↑	UIQM↑	UCIQE↑	NIQE↓
UWCNN[14]	23.68	0.85	4.62	0.56	4.98
ShallowNet[15]	21.64	0.82	4.33	0.51	5.41
WWUIE[16]	24.33	0.86	4.63	0.57	5.13
Deep wave[13]	25.55	0.87	4.53	0.57	5.60
P2CNet[22]	21.71	0.80	**5.42**	0.54	5.54
UWFormer[17]	26.32	**0.89**	4.49	0.57	4.93
DDFormer[18]	22.93	0.86	4.44	**0.61**	5.34
Geo-ViT	**28.62**	0.88	4.48	0.58	**4.63**

Table 7. Quantitative comparison on U45 dataset[27].

Method	UIQM↑	UCIQE↑	NIQE↓
UWCNN[14]	4.29	0.54	4.06
ShallowNet[15]	3.82	0.47	4.17
WWUIE[16]	4.41	0.61	**3.95**
Deep Wave[13]	4.07	0.58	4.02
P2CNet[22]	3.86	0.52	4.33
UWFormer[17]	**4.47**	0.59	4.19
DDFormer[18]	4.38	0.60	5.10
Geo-ViT	4.29	**0.62**	4.06

Table 8. Ablation study on different architectural components evaluated on the UIEB dataset[24].

Method	**Geometry encoder**	**GLCB**	**ViT**	**PSNR (dB)**	**SSIM**
Without hybrid bottleneck(GLCB+ViT)	✓	×	×	21.42	0.90
Without GLCB	✓	×	✓	24.66	0.91
Without ViT	✓	✓	×	20.99	0.89
Without geometry encoder	×	✓	✓	24.68	0.91
Geo-ViT	✓	✓	✓	**25.56**	**0.92**

in successful underwater image restoration. Removal of the GLCB block leads to a drop in PSNR to 24.66 dB and SSIM to 0.91. Similarly, In the absence of the ViT module, there is a more significant drop in accuracy, as a PSNR of 20.99 dB and an SSIM of 0.89 result, which demonstrates the importance of modeling global dependencies when dealing with complicated degradation in an underwater environment. Additionally, by excluding the geometry encoder from the system, the PSNR drops to 24.68 dB while the SSIM becomes 0.91, thereby emphasizing the effectiveness of utilizing geometrical information for

Table 9. Ablation Study of Different Loss Functions on the UIEB Dataset

Loss Configuration	L1	L2	Char.	Perc.	PSNR (dB) / SSIM
L1	✓	×	×	×	24.72 / 0.91
L2	×	✓	×	×	22.33 / 0.89
Char.	×	×	✓	×	24.57 / 0.91
Perc.	×	×	×	✓	24.45 / 0.91
Char.+Perc.	×	×	✓	✓	**25.56 / 0.92**

feature extraction purposes. Among all the possible configurations of the network, the Geo-ViT architecture, which uses all three components, results in the best accuracy values with a PSNR of 25.56 dB and an SSIM of 0.92.

Table 10. Efficiency and Performance Comparison on the UIEB Dataset.

Method	Params (M)↓	FLOPs (G)↓	Time (ms)↓	PSNR (dB)↑
UWCNN [14]	0.59	77.85	39.43	19.84
ShallowNet [15]	0.22	43.26	14.51	24.99
WWUIE [16]	0.73	6.251	17.63	25.38
Deep wave [13]	0.28	18.14	63.7	22.60
P2CNet [22]	2.07	10.36	8.37	18.89
UWFormer [17]	0.95	12.16	25.41	23.62
DDFormer [18]	7.58	4.47	25.84	22.93
Geo-ViT	8.19	18.00	13.10	**25.56**

4.8 Impact of Loss Function Configurations

The effect of various loss functions on the performance of the proposed Geo-ViT network is illustrated in Table 9. It can be seen that the choice of the loss function plays an important role in influencing the results of the underwater image restoration task. Comparing the effects of the individual loss functions, the L1 loss gives PSNR equal to 24.72 dB and SSIM value of 0.91 compared to the L2 loss which gives PSNR and SSIM values of 22.33 dB and 0.89, respectively. Charbonnier loss obtains a PSNR of 24.57 dB and SSIM of 0.91, showing its reliability in dealing with complicated degradation cases while ensuring consistent optimization. Likewise, the perceptual loss attains a PSNR of 24.45 dB and SSIM of 0.91 by virtue of encouraging consistency at the feature level and enhancing perception. Nonetheless, on a standalone basis, it does not yield sufficient accuracy in pixel-level reconstruction. For the model performance, the highest scores were achieved using a combination of the Charbonnier and Perceptual losses, which gave a PSNR score of 25.56 dB and an SSIM score of 0.92. The reason behind such performance enhancement comes from the complimentary nature of these loss functions. The first one allows for retaining high quality at the pixel level, while the latter one provides more realistic appearance in images. Such findings show that the use of the presented hybrid loss is vital for the success of Geo-ViT. The joint minimization of reconstruction and perceptual loss allows achieving a balanced approach to optimization and attains better UIE results than the loss functions used independently from each other.

4.9 Computational Complexity and Efficiency Analysis

The efficiency analysis and performance comparison of our method in the UIEB dataset are provided in Table 10. Our proposed Geo-ViT model attains the best PSNR value of 25.56 dB out of all the compared approaches, indicating that the proposed Geo-ViT model is effective for UIE. More specifically, our model outperforms WWUIE by achieving better PSNR value of 0.18 dB due to structural preservation and global color constancy.

In terms of computation, Geo-ViT includes 8.19M trainable parameters and needs 18.0 GFLOPs, which are more compared to some other lightweight CNN architectures. Nevertheless, the extra parameters contribute to better feature extraction, better structural understanding, and long-range dependencies handling. While having more trainable parameters than some other approaches, the suggested algorithm can perform inference at the speed of only 13.10 ms per one picture, or around 76 FPS. This result outperforms similar models based on transformers like UWFormer and DDFormer in terms of computation.

The proposed approach, named Geo-ViT, is intended to enhance restoration performance while being computationally feasible. While it is true that lightweight models have lower parameters, their performance in enhancing restoration quality is poor compared to the proposed framework. This is because, as stated earlier, lightweight models have poor capacity for global context learning and fail to capture geometric features well.

In summary, the findings show that Geo-ViT provides a favorable trade-off between image restoration quality, model complexity, and computational efficiency. Specifically, the Geo-ViT model outperforms all other models in terms of accuracy, despite providing comparable computational overhead and running time, thereby making it suitable for applications in underwater imaging and marine monitoring.

5 Conclusion

This paper introduced Geo-ViT, an integrated architecture intended to solve the problems of color distortion and light scattering in the underwater environment through the development of a new framework which integrates a Geometry Encoder and hybrid bottleneck. In particular, the geometry encoder is used to preserve the structure and edges of objects by using the differential Sobel operator. On the other hand, the hybrid bottleneck which uses Gated Local Context Block(GLCB) and Vision Transformer(ViT) blocks allows the model to perform texture recovery at local scales while preserving colors at a global scale. The performance evaluation of our network was carried out using five benchmark datasets which include the UIEB, UWSCENES, UWIMAGENET, LSUI, and U45 datasets. Our method has performed better than any existing method based on either CNN or transformer networks in terms of both quantitive and qualitative assessment. For instance, the peak PSNR value on the UIEB dataset has been reported to be 25.56 dB while the SSIM is 0.92. Ablation studies additionally reveal that the combination of the hybrid bottleneck and hybrid loss function plays a crucial role in improving restoration quality while preserving visual realism. Future work will focus on reducing computational complexity and enhancing inference efficiency to facilitate real-time deployment in autonomous underwater vehicles (AUVs) and other resource-constrained underwater imaging systems.

Data Availability

The datasets used in this study are publicly available. The Underwater Image Enhancement Benchmark (UIEB) dataset is available at `https://li-chongyi.github.io/proj_benchmark.html`. The Enhancing Underwater Visual Perception (EUVP) dataset is available at `https://irvlab.cs.umn.edu/resources/euvp-dataset`. The Large-Scale Underwater Image (LSUI) dataset is available at `https://lintaopeng.github.io/code/`. The U45 benchmark dataset is vailable at `https://github.com/IPNUISTlegal/underwater-test-dataset-U45-`.

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Author Contributions

Thallapally Harika contributed to conceptualization, designed the methodology, implemented the model, conducted the experiments, analysed the results, and prepared the original manuscript draft. Kankanala Srinivas supervised the research, reviewed and edited the manuscript, and provided overall guidance throughout the study. All authors reviewed and approved the final manuscript.

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